

KW130 SDR Tuner GUI User Guide

Version: v3.12 Date: 2026-08-26

1. Introduction

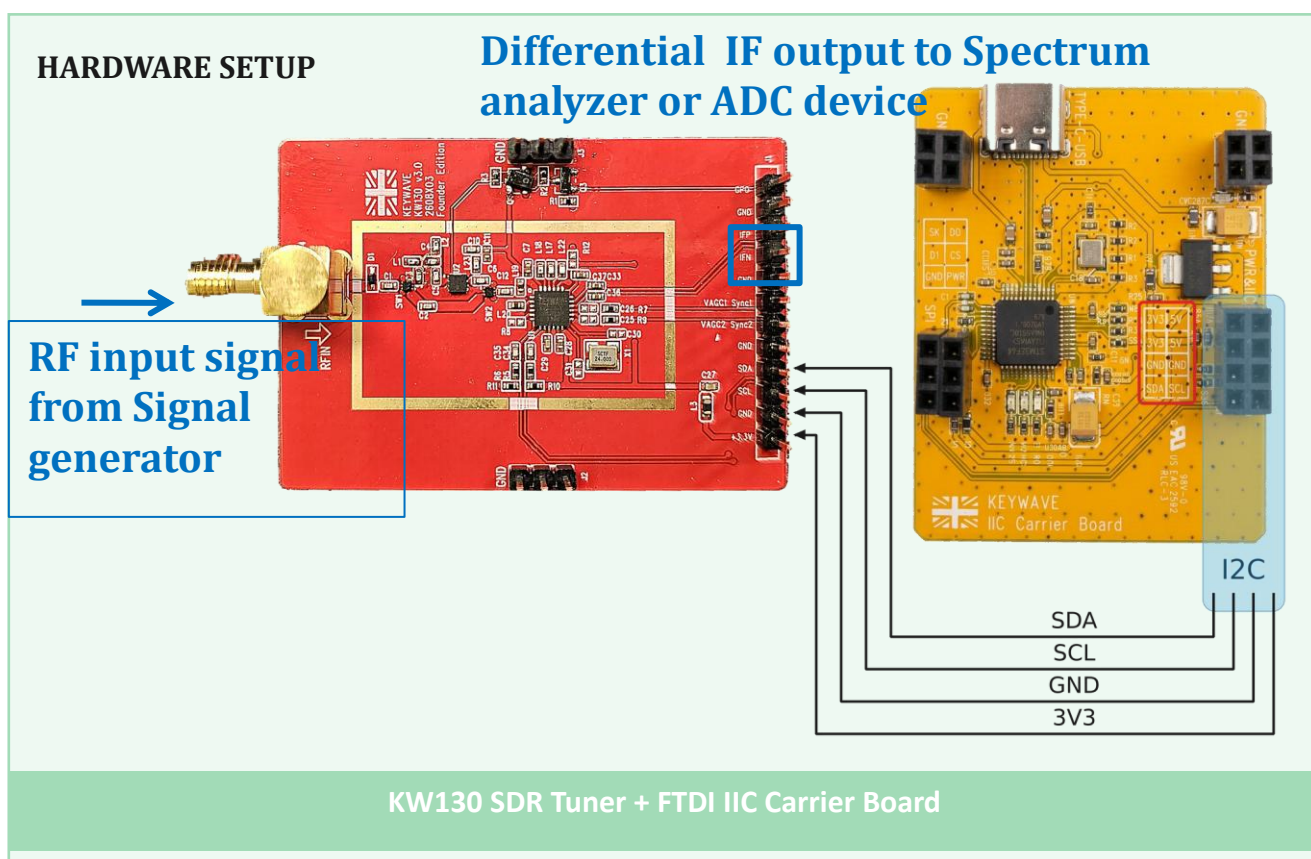
The KW130 GUI application is a configuration and monitoring tool for the KW130 SDR Tuner. This software allows you to:

- Configure tuner parameters
- Evaluated KW130 under different frequency with/without front LNA.

2. System Requirements

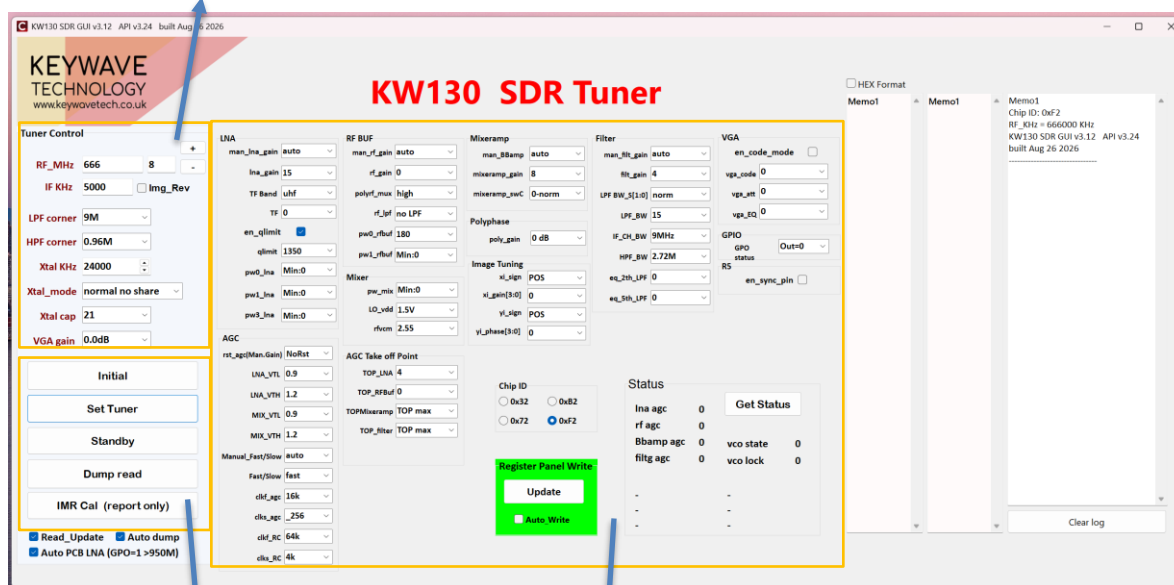
- **Operating System:** Windows 10 or later
- **Hardware:** FT232H USB-to-I²C adapter
- **Driver:** FTDI D2XX driver (FT232H)
- **Connection:** KW130 module connected via I²C

3. Hardware setup



4. GUI Overview

Tuner Input parameter Region,



Normal Operation: Initial -> Set Tuner

Update Parameter if need finetune performance (ex. filter counter course tune)

5. Custom Adjust Settings

Tuner Control on the left holds what defines the channel. It is applied by Initial and by Set Tuner.

The register panel on the right holds the finer settings. Press Update to apply them. AGC, gain, LPF trim and TOP are held across later tunes; anything else on that panel is overwritten by the next Set Tuner.

5.1 Channel

A channel is an IF centre and two filter corners. Set all three:

$$\text{HPF corner} = \text{IF centre} - \text{bandwidth} / 2$$

$$\text{LPF corner} = \text{IF centre} + \text{bandwidth} / 2$$

An 8 MHz channel at a 5 MHz IF is therefore HPF 1 MHz, LPF 9 MHz - pick 0.96M, the nearest HPF step.

IF KHz takes any value; 4570 and 5000 are both ordinary. LPF corner offers 6, 7, 8, 9, 12, 16 and 20 MHz. HPF corner runs 2.72 MHz down to 0.32 MHz in 0.16 MHz steps.

Img_Rev puts the LO below the RF instead of above, which reverses the IF spectrum.

5.2 VGA gain

One step covers the whole range, -24.8 dB to +40.0 dB. Step 9 is 0 dB and is the default. 3.2 dB per step through the middle, 2.4 dB at the two ends.

Step	dB	Step	dB	Step	dB	Step	dB
0	-24.8	6	-9.6	12	+9.6	18	+28.8
1	-22.4	7	-6.4	13	+12.8	19	+32.0
2	-20.0	8	-3.2	14	+16.0	20	+35.2
3	-17.6	9	0.0	15	+19.2	21	+37.6
4	-15.2	10	+3.2	16	+22.4	22	+40.0
5	-12.8	11	+6.4	17	+25.6		

5.3 AGC

Detector window, per path. vth is where the AGC starts reducing gain, vtl where it starts increasing it. Both move in 0.1 V steps: vth 0.4 to 1.9 V, vtl 0.3 to 1.8 V. Defaults are vth 1.3 V and vtl 0.9 V on both paths.

Loop speed, twelve settings from 64 kHz down to 2 Hz, default 4 kHz. The RC corner goes with it - a fast speed and a slow corner cannot be combined, because one selector picks both.

5.4 TOP(AGC)

The level at which each detector calls its block overdriven. Lower it and the AGC backs the gain off sooner, trading sensitivity for headroom.

Detector	Step range	Default
TOP_LNA	0 - 7	4
TOP_RFBuf	0 - 7	1
TOPMixeramp	0 - 7, 0 is the highest TOP	3
TOP_filter	0 - 7, 0 is the highest TOP	4

5.5 Manual gain

Each block follows the AGC or holds a fixed step. In auto the step is stored and takes effect when you switch to manual.

Block	Step range	Default
LNA	0 - 31	auto (agc_control)
RF buffer	0 - 15	
Mixer amp	0 - 11	
Filter	0 - 7	

5.6 LPF fine trim

Moves the LPF corner within the setting chosen in 5.1. 0 is the narrowest, 31 the widest. Each corner has its own default

5.7 Auto PCB LNA

Drives the GPO pin, which on the EVK enables the LNA in front of the tuner. With the box ticked the pin follows the tuned frequency: high above 950 MHz, low at or below it. Standby takes the pin low, so the LNA is off while the tuner is; the next Set Tuner brings it back.

5.8 Image rejection calibration

IMR Cal measures the I/Q imbalance at six frequencies and reports the result. Nothing is applied - you decide whether to use a result, and which one.

The six tones come from dividing the crystal, so where they land depends on it. With 24 MHz they are 864, 608, 480, 320, 160 and 80 MHz; with 27 MHz, 864, 576, 459, 306, 153 and 76.5 MHz. The Status box shows the tone next to each result, which is how you pick the one nearest your frequency.

Results read as (x,y) with a sign per axis, for example 864M (x,y)=(-3,+2). Calibration runs an internal tone through the receiver, so the part cannot receive until the next Set Tuner - press it afterwards whether or not you keep a result.

6. Monitoring

Get Status reads the status registers and refreshes the box:

Field	Meaning
Ina agc	LNA gain the AGC has settled on
rf agc	RF buffer gain
Bbamp agc	Mixer amp gain
filtg agc	Filter gain
vco state	VCO band the PLL selected
vco lock	1 when the PLL is locked
IMR results	the six calibration points, tone and (x,y)

Dump read reads all 64 registers back and refreshes the register panel. Auto dump does it after every Initial, Set Tuner and Standby.

7. Message Area

Every API call reports its result there: Success, or the reason it failed - I2C, PLL did not lock, or a rejected parameter.

8. Troubleshooting

Symptom	Check
"Failed - I2C (KW_I2C_Fail)"	FT232H not plugged in, driver missing, or another program has the adapter open. ftd2xx.dll must sit beside the exe.
"Failed - PLL did not lock (KW_PLL_Unlock)"	RF or IF outside what the LO can reach, or the crystal frequency in the GUI does not match the board.
"Rejected - bad parameter (KW_Bad_Param)"	A value outside its range. The message names the call.
No IF output after IMR Cal or Standby	Both leave the receiver down by design. Press Set Tuner.
A register setting will not stick	Only AGC, gain, LPF trim and TOP are held across a tune. Everything else on the register panel is rewritten by the next Set Tuner.
IMR Cal reports a point that did not finish	The tone PLL did not lock at that frequency, or the engine did not settle. The other points are still usable.